

A Probabilistic Risk Assessment and Risk-Averse Optimization Framework for Sewer Spill Reduction in Wastewater Utilities

Ramesh Manian, Khang Do, Annabelle Jayadinata, Maura Osorio
Sponsor: Chris Foley
Stanford University

Abstract

This paper develops a probabilistic risk assessment (PRA) and risk-averse optimization framework for sanitary sewer spill mitigation in wastewater utilities. The framework integrates historical spill data, severity-category modeling, spill-volume modeling, probabilistic spill occurrence estimation, and operational priors for sparse failure modes. Risk-averse investment allocation is achieved through expected utility maximization with an exponential utility function calibrated to the utility’s risk tolerance, combined with Monte Carlo simulation over stochastic spill outcomes. A case study using 26 years of spill history from the Carmel Area Wastewater District (CAWD) demonstrates how the framework converts operations and maintenance investment decisions into quantified spill-risk reduction outcomes, including reductions in expected annual spill cost and high-severity tail exposure.

1 Introduction

1.1 Motivation

Aging wastewater infrastructure, increasing regulatory scrutiny, climate-driven hydraulic stress, and constrained operational budgets have elevated the importance of quantitative risk-informed decision making in wastewater utilities. Sanitary sewer overflows (SSOs) are unintended releases of untreated wastewater from collection systems prior to reaching a treatment facility. The consequences of SSOs range from localized property damage to large-scale surface-water contamination, beach closures, and regulatory enforcement actions—and they are fundamentally preventable through targeted operations and maintenance (O&M) investment.

The Carmel Area Wastewater District (CAWD) operates approximately 81 miles of gravity and force-main sewer infrastructure serving roughly 11,000 residents on California’s central coast. Wastewater collected from the service area travels by gravity to a wet well, is pumped through headworks for screening and grit removal, and then proceeds through primary clarification, biological

treatment, and tertiary microfiltration and reverse osmosis before reclaimed water is delivered to Pebble Beach golf courses. Any failure in the collection network prior to the headworks constitutes an SSO: untreated wastewater releases directly to the environment without any treatment barrier. This makes collection system integrity the first and most consequential line of defense in CAWD’s environmental protection strategy.

CAWD’s environmental exposure from SSOs is amplified by its geography. The collection system discharges to a treatment plant situated adjacent to the Carmel River lagoon, a coastal water body subject to active hazard monitoring under a Coastal Development Permit from the California Coastal Commission (?). The district’s service area is further bounded by Carmel Bay, designated as an Area of Special Biological Significance under California’s Ocean Plan. Any Category 1 spill that reaches a drainage channel, the river, or the bay carries disproportionate environmental and regulatory consequences relative to a contained spill of the same volume. This geographic reality motivates the severity-weighted cost structure at the core of the framework developed in this paper.

Despite these risk-amplifying conditions, CAWD faces a spill rate higher than comparable utilities and currently lacks a quantitative framework connecting O&M investment decisions to spill-risk reduction outcomes. The district seeks a data-driven plan to minimize the cost of spill consequences under a fixed O&M budget. This paper addresses that objective through three integrated components:

1. A **failure-mode probabilistic risk model** that decomposes annual spill risk into five mutually exclusive failure modes (blockage, pipe failure, pump failure, hydraulic overload, and operational failure), each modeled as a Poisson process with rate calibrated from 26 years of CAWD historical spill records.
2. A **Risk Averse Optimization** that allocates CAWD’s annual O&M budget across four investment levers to maximize expected utility. It allocates to the candidate with the lowest negative expected utility objective.

The consequences of deferred maintenance on aging sewer infrastructure are neither abstract nor distant. In January 2026, a sewer collapse released an estimated 240 to 300 million gallons of untreated sewage into the Potomac River, one of the largest such spills in U.S. history. In April 2026, a federal judge found Mountain View, California in violation of the Clean Water Act and imposed nearly \$1.2 million in civil penalties for allowing raw sewage to enter Stevens Creek. Both cases illustrate the same failure: deferred maintenance on aging infrastructure produces environmental damage, regulatory liability, and remediation costs that dwarf the cost of proactive risk management. For a utility like CAWD, situated adjacent to environmentally sensitive coastal waters, the asymmetry between prevention cost and consequence cost is particularly stark.

1.2 Research Gap

Despite substantial progress in both wastewater asset management and probabilistic risk analysis, existing approaches rarely combine these methods into a single decision-support framework usable by small- and mid-sized utilities. In particular, the literature exhibits the following gaps:

- **Lack of integrated probabilistic risk assessment (PRA) for sewer systems.** Most utility risk studies focus on individual failure modes in isolation rather than decomposing total spill risk into a coherent set of mutually exclusive, exhaustive failure modes with associated occurrence rates, severity distributions, and cost consequences (?).
- **Absence of severity-weighted spill modeling.** Existing benchmarking practices typically report raw spill counts or volumes without weighting by regulatory category or environmental consequence, which obscures the disproportionate contribution of Category 1 surface-water discharges to total risk.
- **No explicit treatment of catastrophic spill events.** Sanitary sewer overflows exhibit heavy-tailed cost distributions, yet most utility planning tools rely on expected-value criteria or deterministic schedules that systematically underweight rare, high-consequence events.
- **Minimal utility-theoretic framing of public infrastructure risk.** Wastewater utilities operate under multi-objective constraints (regulatory compliance, environmental stewardship, public trust, fiscal accountability) that warrant explicit decision-analytic treatment, yet most operational planning tools remain deterministic (?).

This study addresses these gaps by integrating failure-mode-specific PRA, severity-weighted spill modeling,

Monte Carlo simulation, and expected-utility-based risk-averse optimization within a single decision-support framework calibrated to a real wastewater utility.

1.3 Contributions

This paper makes the following contributions:

1. **A normalized probabilistic spill-risk framework** that decomposes annual SSO risk into five mutually exclusive failure modes (blockage, pipe failure, pump failure, hydraulic overload, operational failure) and expresses each as a Poisson process with rate calibrated from historical utility data.
2. **Failure-mode-specific spill modeling** that links operational and maintenance investments to mode-level spill rates through an exponential effectiveness function, capturing diminishing returns on O&M spending.
3. **Severity-category and spill-volume modeling** that translates spill occurrence into regulatory category and discharge volume distributions, enabling explicit modeling of Category 1 surface-water impacts and their disproportionate cost consequences.
4. **Integration of O&M investments with spill-risk reduction** via a Lagrangian convex optimization that allocates a constrained budget across four investment levers (preventive maintenance, workforce capacity, execution reliability, risk-based targeting) to minimize expected annual spill cost.
5. **Risk-averse optimization under catastrophic spill exposure** using a Monte Carlo-expected utility formulation with exponential utility $U(-C) = 1 - e^{C/\rho}$, calibrated to CAWD’s risk tolerance $\rho = \$10\text{M}$, which penalizes high-cost tail outcomes beyond what expected value alone captures.
6. **A real wastewater utility case study** demonstrating the framework using 26 years of spill history from CAWD, with results showing reduction in both expected annual spill cost and high-severity tail exposure under the optimized investment policy.

2 Background and Literature Review

2.1 Wastewater Spill Risk and Utility Management

Sanitary sewer overflows (SSOs) are unintended releases of untreated wastewater from collection systems prior to reaching a treatment facility. The U.S. Environmental Protection Agency estimates that between 23,000

and 75,000 SSO events occur annually in the United States (?), with consequences ranging from localized property damage to large-scale surface-water contamination, beach closures, and regulatory enforcement actions. In California, SSOs are regulated under the State Water Resources Control Board (SWRCB) Statewide General Order, which establishes mandatory reporting requirements and four spill severity categories based on volume, recovery, and surface-water impact. For utilities operating adjacent to designated sensitive receiving waters—such as areas of special biological significance or actively monitored coastal lagoons—the consequence differential between Category 1 surface-water discharges and lower-category contained spills is particularly pronounced, further motivating the severity-weighted risk framework developed in this paper.

Structural and operational failures in sewer systems are inherently multifactorial. ? identify three primary causal domains: construction-related factors (material selection, joint integrity, installation quality), external factors (soil movement, root intrusion, traffic loading), and operational factors (flow velocity, debris accumulation, hydraulic overload). ? unify nine families of deterioration models under a common “bathtub-with-jumps” hazard framework and emphasize the importance of treating right-censoring and survival selection bias when calibrating failure models from historical inspection data. ? extend this perspective through survival analysis with explicit likelihood functions for censored observations.

For mid-sized utilities, ? review 103 publications on sewer failure prediction and conclude that explanatory variables (previous failures, segment length, material, age) exert greater predictive influence than algorithmic choice, and that severe class imbalance (failure rates near 0.1%) complicates direct application of standard classifiers. This finding motivates the operational-prior approach adopted in the present study for sparse failure modes.

The primary physical failure mechanisms observed in the CAWD collection system are summarized in Table 1, together with their dominant drivers, standard mitigation strategies, and the baseline spill rates λ_{k0} (in spills per 100 miles per year) calibrated from the 26-year historical record. Blockage dominates both in frequency and in total system risk, driven by root intrusion into aging vitrified clay pipe (VCP). Pipe failure ranks second in baseline rate and carries a higher expected cost per spill due to its elevated probability of surface-water impact. The three remaining failure modes—pump failure, hydraulic overload, and operational failure—are sparse in the historical record and are assigned rates informed by operational priors, as described in Section 5.2.

As shown in Table 1, **root intrusion** is the dominant failure mode in the CAWD system, reflecting the prevalence of older VCP segments with permeable joints in

proximity to mature trees. Root masses penetrate joint seals, restrict flow, and accumulate debris, producing the high-frequency blockage events that drive annual spill counts. **Structural pipe failure** ranks second in prevalence and arises from material aging, soil movement, and external loading; rehabilitation via CIPP lining and targeted replacement is the standard mitigation. **Fats, oils, and grease (FOG) and debris accumulation** are captured within the blockage mode and driven by upstream commercial and residential discharges, mitigated through FOG inspections and public education. **Pump station failures** occur at moderate frequency and are addressed through mechanical redundancy and SCADA-based condition monitoring. **Hydraulic overload** from inflow and infiltration (I&I) is reported at low frequency in CAWD records, but this likely reflects a data-coding gap rather than true rarity; ? document up to 88% RDI&I reduction following targeted rehabilitation, suggesting hydraulic overload warrants explicit modeling even when underreported. This data gap motivates the operational-prior treatment adopted in Section 5.2.

2.2 Probabilistic Risk Assessment (PRA)

Probabilistic risk assessment (PRA) originated in the nuclear power industry and provides the methodological foundation for this study. ? established the canonical quantitative definition of risk as a set of triplets $\langle S_i, P_i, X_i \rangle$, where S_i is a scenario, P_i is its probability, and X_i is its consequence. This triplet structure—scenario identification, probability estimation, and consequence quantification—maps directly onto the failure-mode decomposition developed in this paper: each failure mode k constitutes a scenario class, its Poisson rate λ_k characterizes its probability, and the severity-weighted cost W_k quantifies its consequence.

PRA has since been extended to civil infrastructure systems, including water and wastewater networks. ? demonstrate the feasibility of applying the PRA framework to networked infrastructure, noting that while PRA is standard practice in nuclear and industrial safety settings, it has not been widely adopted for water and sewer systems despite the analytic tools being directly transferable. Their central finding is that the primary barrier is not methodological but operational: infrastructure analyses routinely omit the likelihood-assessment component of PRA, treating failures as deterministic rather than probabilistic events. The present study addresses this gap directly by calibrating failure-mode Poisson rates from 26 years of CAWD historical spill records and embedding them in a full probabilistic consequence model.

For high-frequency, low-individual-consequence events such as sewer blockages, the Poisson process is the standard occurrence model for pipeline failure frequency es-

Table 1: Primary Failure Mechanisms, CAWD Historical Prevalence, and Baseline Spill Rates (2000–2026)

Failure Mode	Primary Driver	Mitigation Strategy	CAWD Status	λ_{k0}^a
Blockage	VCP joints; root intrusion; FOG	Chemical treatment; hydro-jetting; CCTV inspection	Dominant	6.48
Pipe Failure	Material aging; soil movement	CIPP lining; pipe bursting; targeted replacement	High	0.88
Pump Failure	Mechanical/electrical wear	Redundancy; SCADA monitoring	Moderate	0.049 ^b
Hydraulic Overload	Inflow & Infiltration (I&I)	Smoke testing; man-hole sealing	Low (Data Gap)	0.20 ^b
Operational Failure	Human error; procedure gaps	Training; standard operating procedures	Rare	0.10 ^b

^a Spills per 100 miles per year, calibrated from the 2000–2026 CAWD spill tracking record (system length 77.57 miles used for normalization; see Section 5.1).

^b Rate informed by operational prior due to sparse historical observations; see Section 5.2.

timation (?). Each failure mode k is characterized by an annual rate λ_k , and observed spill counts n_k over an observation horizon T yield maximum-likelihood estimates $\hat{\lambda}_k = n_k/T$. For failure modes where n_k is small or zero, direct maximum-likelihood estimation produces degenerate zero-rate estimates that understate true system risk. Bayesian inference with operationally informed priors avoids this problem by incorporating domain knowledge into the posterior rate distribution, producing stable estimates suitable for downstream simulation even when historical data are sparse (??).

? apply hazard and operability (HAZOP) analysis and fault-tree decomposition to sewer systems, demonstrating that fault-tree logic provides a tractable representation of sewer failure pathways and confirming that structured PRA methods transfer from industrial safety applications to wastewater infrastructure. ? apply simulation-based life-cycle cost analysis to water main rehabilitation decisions, showing how Poisson-rate-based failure models can be integrated with cost consequence functions to produce quantitative investment recommendations. Both studies establish the analytic precedent for the failure-mode PRA framework developed in Section 5, while the present study extends that precedent by adding investment-dependent failure rates, severity-weighted cost modeling, and risk-averse optimization under budget constraints—components absent from prior infrastructure PRA applications.

2.3 Decision Analysis and Risk Aversion

Classical decision analysis under uncertainty selects among alternatives by maximizing expected utility (??). Under risk neutrality, utility is linear in cost and the decision criterion reduces to expected-value minimization.

Under risk aversion, utility is concave in cost and the decision criterion penalizes downside exposure beyond what expected value alone captures.

The exponential utility function provides a tractable and widely used representation of risk-averse preferences. For a cost outcome C , the exponential utility function takes the form:

$$U(-C) = 1 - e^{C/\rho}$$

where $\rho > 0$ is the risk tolerance parameter. Smaller values of ρ correspond to greater risk aversion: the decision maker increasingly penalizes high-cost outcomes relative to their expected value. As $\rho \rightarrow \infty$, the exponential utility function approaches risk neutrality and the decision criterion converges to expected-cost minimization. This shifted form is equivalent to $U(-C) = -e^{C/\rho}$ for optimization purposes since adding a constant does not change the argmax; the form $1 - e^{C/\rho}$ is used throughout this study because it bounds utility from above by 1 (achieved only at zero cost) and avoids numerical overflow in Monte Carlo simulation over large scenario sets.

For CAWD, stakeholder input and budget context yield a risk tolerance of $\rho = \$10,000,000$, equal to ten times the annual collections O&M budget of \$1M (?).

Under this framework, the optimal investment policy maximizes expected utility over the stochastic spill cost distribution:

$$\max_I E[U(-C(I, \omega))] = \max_I E[1 - e^{C(I, \omega)/\rho}]$$

where $C(I, \omega)$ is the realized annual spill cost under investment allocation I and stochastic scenario ω . Because the O&M budget B is fixed and fully allocated regardless of how it is distributed across investment levers, the investment cost itself is not uncertain. All cost uncertainty enters through the spill consequence side, which

makes expected utility maximization over spill outcomes the appropriate decision criterion.

The severity-weighted cost structure of the CAWD consequence model, in which Category 1 surface-water spills carry per-gallon fines of up to \$10 per gallon on top of fixed O&M response costs, already encodes a degree of consequence weighting. The exponential utility layer further penalizes variance and tail exposure beyond what those severity weights alone capture, producing a framework that is sensitive to the full distribution of annual spill cost rather than only its mean.

2.4 Infrastructure Investment Optimization

Budget-constrained infrastructure investment optimization has been studied extensively in the context of water and wastewater asset management. ? develop a decision support framework for sewer network renewal planning that links rehabilitation expenditures directly to failure-rate outcomes, demonstrating that quantitative risk-based allocation outperforms reactive or schedule-based approaches under budget constraints. ? establish the analytical basis for treating infrastructure asset renewal as a scheduling problem under uncertainty, modeling asset deterioration as a semi-Markov process and optimizing inspection and renewal timing to minimize expected life-cycle cost.

Risk-based targeting—directing inspection and maintenance resources toward the highest-risk assets rather than cycling through the system uniformly—is a central principle in modern infrastructure asset management. ? documents condition-based maintenance strategies for sewer networks and shows that targeting maintenance toward assets with the highest deterioration rates and blockage risk reduces system-wide failure frequency more efficiently per dollar spent than time-based uniform schedules. ? further demonstrate that blockage rates across utility systems are systematically lower in utilities that use active, condition-informed maintenance programs, providing empirical support for the link between maintenance targeting and failure-rate reduction. This finding motivates the pipe-level logistic regression model developed in Section 4.2, which produces a Combined Risk Index that operationalizes risk-based targeting as Investment Lever I_4 .

The exponential effectiveness structure adopted in this study, $\lambda_k(I) = \lambda_k^0 \exp(-\sum_i \alpha_{ki} I_i)$, reflects the principle of diminishing returns to preventive maintenance investment, adopted as a standard modeling assumption in infrastructure maintenance optimization (??). The exponential form captures this property analytically: each additional unit of investment reduces the failure rate by a constant fraction rather than a constant absolute amount, so the marginal benefit of investment declines as the rate approaches its lower bound. The Lagrangian op-

timization framework developed in Section 7 formalizes the budget-constrained allocation problem for CAWD, with the effectiveness matrix α_{ki} serving as the quantitative bridge between investment decisions and failure-mode risk reduction.

3 System Description and Problem Formulation

3.1 Carmel Area Wastewater District

The Carmel Area Wastewater District (CAWD) is a public wastewater utility serving approximately 11,000 residents in the Carmel-by-the-Sea region on California’s central coast. CAWD treats and manages roughly 1.3 million gallons of wastewater per day. The collection system comprises approximately 1,807 individual pipe segments totaling 81 miles of gravity and force-main sewer infrastructure, of which 77.57 miles of gravity sewer are used for failure-rate normalization throughout this study (see Section 5.1). The system serves a mixed residential, commercial, and hospitality customer base, with significant seasonal variation in flow driven by tourism.

The geographic extent of the collection network is shown in Figure 1. The system has a dense urban core serving Carmel-by-the-Sea, with force main runs extending east toward Pebble Beach and southeast toward the treatment plant situated at the mouth of the Carmel River. This geographic dispersion — with long lateral runs connecting low-density residential areas to the main collection network — means that blockage risk is not uniformly distributed across the system, and that pipe-level risk targeting has material operational value.

The district’s operational environment presents three risk-amplifying characteristics. First, proximity to the Pacific Ocean and several environmentally sensitive watersheds, including the Carmel River and Carmel Bay Area of Special Biological Significance, elevates the consequence of any Category 1 surface-water spill. The treatment plant sits adjacent to the Carmel River lagoon, a coastal water body subject to active hazard monitoring under a Coastal Development Permit from the California Coastal Commission (?). Second, a substantial fraction of the collection network consists of aging vitrified clay pipe (VCP) installed prior to 1970, which exhibits elevated susceptibility to root intrusion and joint failure. Third, as a small utility, CAWD operates under a constrained operations and maintenance budget that must be allocated across competing priorities.

Over the 26-year historical record (2000–2026), CAWD has recorded spill events spanning all four SWRCB severity categories, with annual spill counts that exceed benchmarking norms for utilities of comparable size. This excess spill frequency motivates the present study

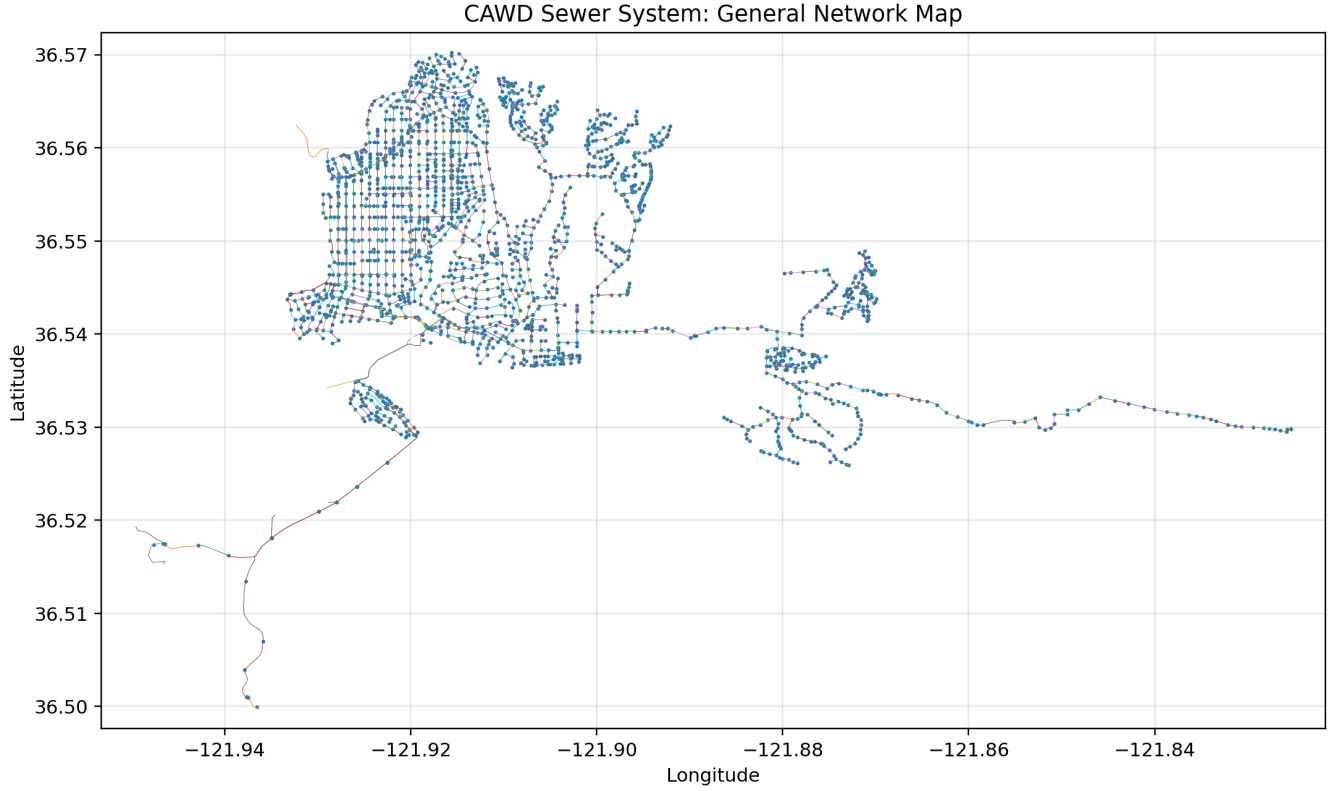


Figure 1: CAWD sewer collection system network. The dense core serves Carmel-by-the-Sea, with force main runs extending east toward Pebble Beach and southeast toward the treatment plant at the mouth of the Carmel River. The system comprises approximately 1,807 pipe segments across 77.57 miles of gravity sewer and associated force-main infrastructure.

and provides the empirical basis for failure-mode rate calibration in Section 5.

3.2 Failure Modes

Annual spill events are decomposed into five mutually exclusive and collectively exhaustive failure modes:

- B = Blockage (root intrusion, FOG, debris),
- F = Pipe failure (structural, joint, corrosion),
- P = Pump failure (mechanical, electrical, control),
- H = Hydraulic overload (I&I, wet-weather capacity),
- O = Operational failure (human error, procedure).

Each mode is modeled as an independent Poisson process with rate λ_k calibrated from historical data (Section 5.1) augmented by operational priors for sparse modes (Section 5.2).

3.3 Spill Severity Categories

Spills are classified into four severity categories following California’s Statewide General Waste Discharge Re-

quirements:

- **Category 1** is the most consequential: any spill that reaches surface water, a drainage channel, or a municipal storm sewer regardless of volume. These events carry the highest regulatory, environmental, and reputational exposure and are the primary driver of tail-risk in the cost distribution.
- **Category 2** covers spills of 1,000 gallons or greater that do not reach surface water and are fully recovered. These trigger mandatory state reporting but produce lower environmental consequence than Category 1.
- **Category 3** covers spills under 1,000 gallons that do not reach surface water and are fully recovered. State reporting is required but regulatory exposure is minimal.
- **Category 4** is a CAWD-specific subcategory capturing spills under 50 gallons—events too minor to drive material cost but meaningful for tracking maintenance effectiveness and identifying early-stage failure signals.

Given failure mode k , the conditional category probability is denoted $q_{kc} = P(C = c \mid k)$, with $\sum_c q_{kc} =$

1. These conditional probabilities are estimated from historical CAWD records and applied during Monte Carlo simulation (Section 9) to translate mode-level spill counts into category-level spill outcomes.

3.4 Decision Variables

The decision vector $I = (I_1, I_2, I_3, I_4)$ represents annual O&M investment (in units of \$100,000) across four management levers:

- **I_1 : Preventive Maintenance.** Routine cleaning, hydro-jetting, root treatment, and CCTV inspection. Primary effect: blockage rate reduction.
- **I_2 : Workforce Capacity.** Crew size, overtime availability, and emergency-response readiness. Primary effect: reduced spill duration and improved operational reliability.
- **I_3 : Execution Reliability.** Training, standard operating procedures, quality assurance. Primary effect: operational-failure rate reduction.
- **I_4 : Risk-Based Targeting.** Condition assessment, data analytics, and pipe-level risk scoring to direct I_1 toward highest-risk assets. Primary effect: multiplicative enhancement of preventive-maintenance effectiveness.

The investments are constrained by the annual O&M budget B : $\sum_i I_i \leq B$, with $I_i \geq 0$ for all i . Capital expenditures such as pipe replacement, CIPP lining programs, and major pump station rehabilitation are appropriated through a separate multi-year Capital Improvement Program and are outside the scope of the O&M allocation problem studied here (?). The mapping from I to failure-mode rates $\lambda_k(I)$ is developed in Section 6.

4 Data & Methods

4.1 Data Sources

This study draws on three main data sources from CAWD.

The first and most detailed source is the V&A Consulting Engineers Data Science Services report. From this report, an integrated dataset was constructed linking pipe characteristics—diameter, length, material, installation year, and drainage basin—with cleaning history and observations recorded prior to each cleaning event.

The second source is CAWD’s 2025 Annual Collections Report (?), which provides current system context including 86 miles of gravity sewer and the 2025 spill summary. A 10-year cause-of-spill analysis (2014–2024) published in this report shows that root intrusion accounts

for 65% of recorded spills—combining direct root blockage (56%) and free-flowing roots (9%)—while grit and grease blockage accounts for 27%, and all remaining external causes account for 8%. These proportions serve as the empirical basis for the blockage-cause weights in the Combined Risk Index.

The third source is CAWD’s Sewage Spill Tracking record, which covers February 2000 through August 2024 and includes 145 spill events with information on date, location, estimated volume, and cause. This record provides the empirical basis for failure-mode rate calibration and spill severity classification.

A representative sample of the integrated dataset is provided in Appendix C.

4.2 Probabilistic Risk Assessment Framework

This section develops the probabilistic risk assessment (PRA) foundation of the study. Following the classical PRA formulation of Kaplan and Garrick (1981), risk is represented through three elements: the failure scenario, its probability of occurrence, and its associated consequence. In the context of sanitary sewer overflows, the scenarios correspond to the five mutually exclusive failure modes defined in Section 3.2, the probabilities are represented by annual spill occurrence rates, and the consequences are captured through spill severity categories, discharge volumes, and resulting costs.

The objective of this section is to translate historical spill records into a quantitative risk model that can support both simulation and optimization. The resulting framework provides the probabilistic inputs required for the investment-dependent risk reduction model developed in Section 6 and the optimization framework developed in Sections 7 and 8.

4.2.1 Baseline Spill Rate Estimation

The first step in the PRA is the estimation of baseline failure frequencies. Historical spill records provide observations of realized failures over a finite period of operation. These observations are converted into annualized failure rates that can be compared across systems of different size and used directly in probabilistic occurrence models.

Because wastewater utilities commonly benchmark performance using spills per 100 miles of sewer per year, raw spill counts must be normalized by both observation time and system length. The resulting baseline rate represents the expected annual frequency of spills attributable to a given failure mode prior to any additional investment intervention.

The baseline spill rate for failure mode k is estimated as:

$$\lambda_k^0 = \frac{100}{L} \frac{n_k}{T}$$

where L is system length in miles, T is the observation horizon in years, and n_k is the observed spill count for mode k . Normalizing by system length produces a rate expressed as spills per 100 miles per year, enabling comparison against industry benchmarks.

4.2.2 Sparse-Mode Operational Priors

Not all failure modes are equally represented in historical records. Some events, such as pump failures, hydraulic overloads, and operational failures, occur infrequently and may be underreported due to data-coding practices or limited historical observation periods. Direct estimation from sparse observations can therefore underestimate true risk and may even produce zero-rate estimates when no events have been observed.

To avoid this problem, sparse failure modes are supplemented with operational priors derived from engineering judgment, utility operating experience, and published industry observations. These priors provide conservative but realistic estimates of baseline risk and ensure that potentially important failure mechanisms remain represented within the overall risk model.

4.2.3 Spill Occurrence Model

Once baseline failure rates have been established, a stochastic model is required to represent annual spill occurrence. Sewer spills are discrete events occurring randomly through time and are naturally modeled as counting processes. The Poisson process is widely used in reliability engineering and infrastructure risk analysis because it provides a tractable representation of event frequencies when individual events occur independently and relatively infrequently.

The Poisson formulation allows annual spill counts to be simulated directly and provides the probabilistic link between failure-mode rates and system-level spill occurrence.

For each failure mode, annual spill count is modeled as:

$$N_k \sim \text{Poisson}(\lambda_k).$$

Total annual spill occurrence is:

$$N = \sum_k N_k.$$

4.2.4 Severity and Volume Modeling

Spill occurrence alone does not fully characterize risk. Two spills occurring through the same failure mechanism

can differ dramatically in environmental consequence depending on the volume released and whether the spill reaches surface water. Consequently, spill severity must be modeled separately from spill occurrence.

This study represents severity through conditional category probabilities and associated spill-volume distributions. Together, these distributions convert occurrence frequency into consequence outcomes that can be translated into economic and regulatory impact.

Given failure mode k , the conditional category probability is:

$$q_{kc} = P(C = c \mid k)$$

Spill volume is modeled through the conditional distribution $V \mid (k, c)$, which is estimated empirically from historical CAWD records.

4.2.5 Spill Cost Model

The final component of the PRA framework is consequence quantification. Regulatory agencies, utility managers, and governing boards ultimately evaluate spill performance in terms of operational, environmental, and financial consequences. The spill cost model therefore converts simulated spill volumes and severity categories into monetary outcomes that can be compared across scenarios and incorporated into optimization.

This consequence model serves as the bridge between physical spill events and decision analysis. It provides the objective function inputs required for both expected-cost minimization and risk-averse optimization.

Total spill cost is a function of spill volume and severity category:

$$C(V, S)$$

where V is spill volume and S is the spill severity category.

5 Investment-Dependent Failure Modeling

The baseline risk model developed in Section 5 characterizes spill risk under existing operating conditions. However, the primary objective of this study is not merely to estimate risk but to determine how risk changes as a function of operational and maintenance investment decisions. This requires an explicit model linking management actions to failure-mode frequencies.

The framework developed in this section establishes that connection. Investment decisions influence operator actions, operator actions influence failure mechanisms, and failure mechanisms determine spill occurrence. The resulting formulation enables quantitative evaluation of al-

ternative budget allocations and provides the analytical foundation for optimization.

5.1 SAM-Based Risk Reduction Structure

The causal chain linking management decisions to spill outcomes follows a Systems Analysis and Management (SAM) structure (?):

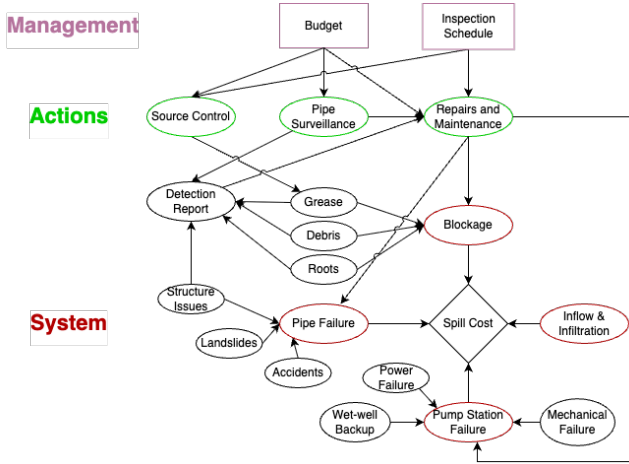


Figure 2: SAM Model for Sewage Spill Risk Analysis

The Systems Analysis and Management framework provides a structured representation of how management decisions propagate through an operational system. Rather than treating spill occurrence as an exogenous process, SAM explicitly recognizes that management actions influence operational behavior, which in turn influences system performance.

Within the wastewater context, investments in improvement of maintenance frequency, increasing maintenance staff, training & SOP, and risk-based targeting affect the likelihood of specific failure modes occurring. Representing these relationships explicitly clarifies the causal pathway between spending decisions and risk reduction outcomes.

5.2 Effectiveness Matrix

The effectiveness of investment lever i on failure mode k is captured by the coefficient α_{ki} , where larger values indicate greater marginal risk reduction per unit of investment.

Not all investments affect all failure modes equally. Preventive maintenance primarily reduces blockage risk, while training and execution reliability primarily influence operational failures. To capture these heterogeneous effects, the framework introduces an effectiveness matrix whose elements quantify the marginal impact of each investment lever on each failure mode.

The effectiveness matrix serves as the quantitative mechanism by which management decisions alter baseline risk. It therefore forms the central linkage between operations and maintenance expenditures and the probabilistic spill model.

5.3 Investment-Dependent Spill Rates

The effectiveness matrix must be translated into a functional relationship between investment and failure frequency. This study adopts an exponential effectiveness formulation, which is widely used in reliability and maintenance optimization because it naturally captures diminishing returns.

Under this representation, early investments generate the largest reductions in failure rate, while additional spending produces progressively smaller incremental benefits. This behavior is consistent with practical utility operations, where major deficiencies are addressed first and remaining improvements become increasingly difficult to achieve.

The investment-dependent spill rate for failure mode k is:

$$\lambda_k(I) = \lambda_k^0 \exp \left(- \sum_i \alpha_{ki} I_i \right)$$

This exponential form captures diminishing returns on O&M spending: each additional unit of investment reduces the spill rate by a constant fraction rather than a constant absolute amount, which is consistent with the empirical literature on preventive maintenance effectiveness.

6 Risk-Averse Optimization Framework

6.1 Motivation for Risk Aversion

Although expected-value optimization minimizes average annual spill cost, wastewater spill risk is inherently heavy-tailed. Most spill events are relatively small and contained, while a small number of catastrophic surface-water discharges produce disproportionately large environmental, regulatory, and financial consequences. As a public wastewater utility operating within environmentally sensitive watersheds, CAWD must consider not only average operational performance but also exposure to rare high-consequence spill events.

Risk-neutral optimization frameworks may therefore underestimate catastrophic spill exposure by compressing spill uncertainty into deterministic expected-cost terms. In practice, wastewater infrastructure governance is inherently risk-averse because utilities must maintain regu-

latory compliance, environmental stewardship, and public trust in addition to minimizing operational expenditure.

6.2 Monte Carlo Spill-Cost Simulation

Annual spill outcomes are inherently uncertain because spill occurrence, severity, and volume vary from year to year. For a fixed investment allocation I , the resulting annual spill cost

$$C(I, \omega)$$

is therefore a random variable, where ω denotes a realized spill scenario.

Monte Carlo simulation is used to characterize the distribution of annual spill costs. Repeated simulation generates a large sample of possible annual outcomes, allowing estimation of the probability distribution of annual spill consequences.

The simulation framework preserves uncertainty in three components:

1. spill occurrence,
2. spill severity category,
3. spill volume.

Together, these sources of uncertainty produce an empirical distribution of annual spill costs

$$f_{C(I)}(c).$$

This distribution serves as the input to the risk-averse decision framework developed in the next section.

6.3 Risk-Averse Optimization

The risk-averse optimization problem is formulated using expected utility maximization rather than a CVaR penalty. For each investment allocation I , Monte Carlo simulation produces a distribution of annual spill costs $C(I, \omega)$. The decision maker evaluates this distribution using the exponential utility function

$$U(-C) = 1 - e^{C/\rho},$$

where ρ is the risk tolerance parameter. Smaller values of ρ imply greater risk aversion because high-cost outcomes receive disproportionately lower utility.

The optimization problem is therefore:

$$\max_I E[U(-C(I, \omega))]$$

subject to

$$\sum_i I_i \leq B, \quad I_i \geq 0.$$

Equivalently, the computational implementation minimizes the negative expected utility:

$$\min_I -E[U(-C(I, \omega))].$$

In this study, $\rho = \$10\text{M}$ is used, representing relatively high risk tolerance. For each candidate allocation, expected utility is estimated from simulated annual spill costs. The best allocation is the candidate with the lowest negative expected utility objective. This formulation penalizes high-cost tail outcomes through the curvature of the utility function rather than through an explicit CVaR term.

6.4 Stochastic Dominance Comparisons

Alternative investment policies may additionally be compared using stochastic dominance criteria. First-order stochastic dominance (FSD) occurs when one policy produces lower spill costs across all outcome levels, while second-order stochastic dominance (SSD) incorporates risk-sensitive preferences by favoring policies with reduced downside variability and catastrophic spill exposure.

Stochastic dominance analysis provides an additional mechanism for comparing investment strategies beyond expected-value metrics alone and supports robust policy comparison under uncertainty.

6.5 Literature Basis

The risk-averse optimization framework developed in this section rests on three well-established pillars of decision analysis.

The first is the axiomatic foundation of expected utility theory. ? establishes that a rational decision maker facing uncertain outcomes should maximize expected utility, where the utility function encodes the decision maker's preferences over outcome distributions rather than just their mean. The exponential utility function $U(-C) = 1 - e^{C/\rho}$ used in this study satisfies the constant absolute risk aversion (CARA) property, meaning the decision maker's risk premium for a given gamble does not depend on current wealth. CARA is a standard and tractable assumption in infrastructure risk management contexts where the decision maker—a public utility board—evaluates risk exposure in terms of budget-relative consequence rather than absolute wealth level. The risk tolerance parameter $\rho = \$10\text{M}$ is calibrated to be ten times CAWD's annual O&M budget, giving it a concrete operational interpretation: the board is willing

to pay up to one annual budget in certain-equivalent cost to eliminate the variance of a catastrophic spill year.

The second pillar is multi-attribute utility theory. ? extend the expected utility framework to decisions involving multiple conflicting objectives—in the present case, minimizing expected annual spill cost, minimizing regulatory exposure, and managing tail-risk—and provide the theoretical basis for collapsing these into a single scalar utility criterion. The severity-weighted cost function $C(I, \omega) = \sum_k \sum_{j=1}^{N_k(\omega)} C_{k,j,\omega}$ defined in Section 8.2 performs this collapse operationally: regulatory category, spill volume, and surface-water impact are all translated into a single dollar-denominated consequence measure before the utility function is applied. This preserves analytical tractability while retaining the multi-dimensional consequence structure of the CAWD cost model.

The third pillar is the use of Monte Carlo simulation to evaluate expected utility under stochastic spill outcomes. Because the annual spill cost $C(I, \omega)$ is a random variable whose distribution cannot be expressed in closed form under the full severity-category and volume model, direct analytical maximization of $E[U(-C(I, \omega))]$ is intractable. Monte Carlo simulation with 50,000 replications provides a stable empirical approximation of the expected utility surface across investment allocations, enabling numerical optimization over the budget simplex. This simulation-based approach to expected utility maximization is standard practice in infrastructure and environmental risk management applications where consequence distributions are complex and analytically inconvenient (?).

Together, these three pillars—expected utility theory, multi-attribute consequence modeling, and Monte Carlo simulation—form a coherent and well-grounded methodological basis for the risk-averse optimization framework. The practical output, a probability statement that the optimized policy reduces $P(\text{Cost} > \$1\text{M})$ from 17.7% to 9.1%, is a direct product of this distributional treatment and would not be available under a deterministic or expected-value-only framework.

7 Monte Carlo Simulation Framework

7.1 Simulation Configuration

Monte Carlo simulation was performed using 50,000 simulated annual spill scenarios for each policy evaluation. Candidate investment allocations were generated subject to the budget constraint $\sum_i I_i \leq B$. Candidate policies were initially screened using lower-fidelity simulations, after which the highest-performing allocations were re-evaluated using full Monte Carlo simulation to estimate expected annual cost, P90 and P95 spill cost, probability

of annual cost exceeding \$1M, and expected utility.

Simulation inputs grounded in real CAWD data include the baseline failure rates λ_{k0} , the category probabilities q_{kc} for blockage and pipe failure, and the historical spill volume arrays. Rates and category probabilities for the three sparse failure modes (pump failure, hydraulic overload, operational failure) are assigned using operational priors as described in Section 5.2. Simulation outputs were used to compare the baseline no-investment condition against the optimized \$1M preventive-maintenance allocation under the risk-averse expected utility criterion.

7.2 Simulation Algorithm

For each candidate investment allocation I , the simulation first computes investment-adjusted failure rates for each failure mode:

$$\lambda_k(I) = \lambda_k^0 \exp \left(- \sum_i \alpha_{ki} I_i \right).$$

Using these adjusted failure rates, annual spill outcomes are generated through Monte Carlo simulation. For each simulated year:

1. For each failure mode k , sample the annual spill count

$$N_k \sim \text{Poisson}(\lambda_k(I)).$$

2. For each simulated spill event:

- (a) Sample a spill severity category using the conditional category probabilities q_{kc} .
- (b) Sample a spill volume from the corresponding empirical or fitted volume distribution.
- (c) Compute spill consequence cost using the spill-cost model developed in Section 6.

3. Aggregate spill-event costs to obtain total annual spill cost

$$C(I, \omega) = \sum_{j=1}^N C_j,$$

where

$$N = \sum_k N_k$$

is the total number of spill events occurring during the simulated year.

4. Record annual performance measures including total spill count, spill categories, spill volume, and annual spill cost.

The procedure is repeated for 50,000 simulated years for each candidate investment allocation. The resulting empirical distribution of annual spill costs is then used to estimate expected annual cost, percentile risk measures, probability of catastrophic spill years, and expected utility under the risk-averse optimization framework.

Appendix B provides pseudocode for the complete simulation procedure.

7.3 Scenario Analysis

The random search optimization evaluates 305 candidate allocations in total: five deterministic candidates covering the four pure corner allocations ($I = [10, 0, 0, 0]$, $[0, 10, 0, 0]$, $[0, 0, 10, 0]$, $[0, 0, 0, 10]$) and the uniform split ($I = [2.5, 2.5, 2.5, 2.5]$), augmented by 300 allocations generated by randomly partitioning the \$1M budget across the four investment levers, ensuring the budget constraint $\sum_i I_i = B$ is satisfied by construction. Each candidate is first evaluated at a screening resolution of 2,000 simulated years using the negative expected utility objective $-E[U(-C(I, \omega))]$. The ten lowest-objective candidates from screening are then re-evaluated at full resolution using 50,000 simulated years to confirm the optimal allocation. The top ten candidates from the screening stage are reported in Table 2.

Table 2: Top Ten Candidate Allocations from Random Search (Screening Stage, 2,000 Scenarios)

I_1	I_2	I_3	I_4	Neg. EU Objective
9.598	0.016	0.023	0.363	0.053753
10.000	0.000	0.000	0.000	0.054201
8.133	0.107	0.567	1.194	0.054569
6.893	0.120	0.317	2.671	0.054644
2.460	0.177	0.096	7.266	0.054725
5.545	0.418	0.408	3.629	0.054945
7.707	0.103	1.368	0.821	0.055185
8.551	0.125	0.317	1.007	0.055283
6.282	0.914	0.043	2.762	0.055342
7.676	0.802	0.999	0.523	0.055455

Several patterns in Table 2 are noteworthy. First, all ten top candidates allocate the majority of their budget to I_1 (preventive maintenance), with I_1 ranging from 2.460 to 10.000 units across the top ten. No candidate that allocates the majority of its budget to I_2 , I_3 , or I_4 alone appears in the top ten, confirming that preventive maintenance is the dominant lever under the current effectiveness matrix and spill-rate structure. Second, the best non-corner candidate at the screening stage is $I = [9.598, 0.016, 0.023, 0.363]$ with an objective of 0.053753, slightly outperforming the pure corner $[10, 0, 0, 0]$ at 0.054201. This apparent advantage, however, is an artifact of simulation noise at the 2,000-

scenario screening resolution.

Upon re-evaluation at full resolution (50,000 scenarios), the pure corner allocation $I^* = [10, 0, 0, 0]$ is confirmed as optimal with a negative expected utility objective of 0.054592. The near-miss candidate and all other mixed allocations lose their apparent advantage at higher resolution, consistent with the interpretation that the marginal benefit of diverting any budget away from preventive maintenance is negligible and within the simulation noise of the screening stage. This result is robust to the random seed used in screening: the corner allocation $[10, 0, 0, 0]$ appears in the top two candidates regardless of seed, indicating that its dominance is not an artifact of the particular random candidates generated.

The cost distributions for the baseline and optimized policies across 50,000 simulated years are compared in Figure 3. The optimized distribution is shifted substantially leftward relative to the baseline, with the modal annual cost declining from approximately \$500,000–\$600,000 under the baseline to approximately \$350,000–\$450,000 under the optimized allocation. The right tail of the distribution is also compressed: the probability mass above \$1M declines from 17.7% to 9.1%, and the P95 cost declines from \$1,444,638 to \$1,214,527. The shape of both distributions is right-skewed, reflecting the occasional occurrence of high-severity spill events that produce costs well above the mean. This skewness is precisely the feature that motivates the risk-averse exponential utility framework: the expected-cost criterion would treat the right tail symmetrically with the left tail, whereas the exponential utility criterion explicitly down-weights allocations that leave the right tail heavy.

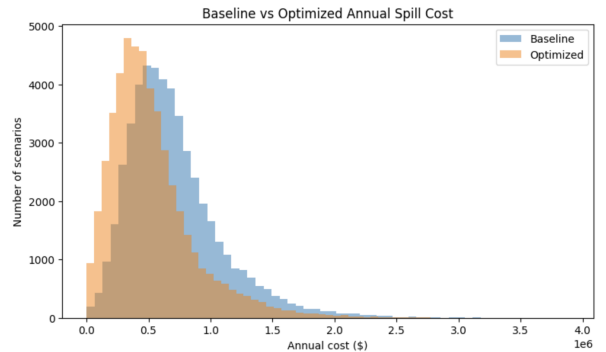


Figure 3: Simulated annual spill cost distributions for the baseline (no investment) and optimized (\$1M preventive maintenance) policies across 50,000 scenarios. The optimized distribution shifts leftward and compresses the right tail relative to the baseline.

The scenario analysis further confirms that the investment landscape is not flat near the optimum. The gap between the best candidate objective (0.053753 at screening, 0.054592 at full resolution) and the worst top-ten candidate (0.055455) represents a meaningful differ-

ence in expected utility, equivalent to a difference of several percentage points in $P(\text{Cost} > \$1\text{M})$. Allocations that place significant weight on I_2 (workforce capacity) or I_3 (execution reliability) in isolation perform substantially worse, because those levers act primarily on pump failure and operational failure — the two lowest-frequency modes in the CAWD baseline. Risk-based targeting (I_4) shows non-trivial performance in several top-ten candidates, appearing as a secondary lever alongside a large I_1 allocation, but does not displace preventive maintenance as the primary driver of risk reduction under the current parameter estimates.

8 Case Study Results

8.1 Historical Spill Reclassification

The 26-year CAWD spill tracking record covers February 2000 through August 2024 and contains 145 documented spill events. Each event was reviewed and assigned to one of the five mutually exclusive failure modes defined in Section 3.2. Classification was based on the cause description recorded in the tracking workbook, cross-referenced against the operational definitions established in Section 3.2. Events with ambiguous or compound cause descriptions were assigned to the mode most directly responsible for the initial system failure.

The classification yielded 132 blockage events, 19 pipe failure events, and 1 operational failure event over the observation period. No pump failure or hydraulic overload events were identified in the labeled records, reflecting either true rarity or systematic under-coding of these modes in the tracking workbook. The three data-sparse modes are treated with operational priors as described in Section 5.2. The blockage mode dominates the historical record, accounting for approximately 87% of all classified events, consistent with CAWD’s 10-year cause-of-spill analysis showing root intrusion and grease blockage as the leading failure drivers (?).

8.2 Baseline Spill Rates

Baseline failure-mode rates are estimated using the normalized rate formula:

$$\lambda_k^0 = \frac{100}{L} \cdot \frac{n_k}{T}$$

where $L = 77.57$ miles is the gravity sewer system length used for normalization, T is the observation horizon in years, and n_k is the classified spill count for mode k . The resulting baseline rates, severity weights, and annual expected costs are reported in Table 3.

Table 3: Baseline Failure Rates, Severity Weights, and Annual Expected Spill Cost

Failure Mode	λ_{k0}	W_k	Annual Cost
Blockage	6.483	\$76,633	\$245,226
Pipe Failure	0.884	\$129,360	\$206,976
Pump Failure	0.049	\$86,763	\$52,058
Hydraulic Overload	0.200	\$105,300	\$21,060
Operational Failure	0.100	\$218,246	\$43,649
Total	7.715	—	\$568,968

The total baseline rate of 7.715 spills per 100 miles per year implies a near-certain probability of at least one spill annually ($P(N \geq 1) = 99.7\%$) and a greater-than-even probability of more than five spills in a given year ($P(N > 5) = 52.2\%$). Blockage dominates both in frequency and in absolute annual expected cost, contributing \$245,226 of the \$568,968 total baseline expected annual spill cost. Despite its lower frequency, pipe failure contributes \$206,976 due to its elevated expected cost per spill, which reflects a higher probability of surface-water impact relative to blockage events. Operational failure carries the highest expected cost per spill (\$218,246) because its severity prior places greater mass on Category 1 and Category 2 outcomes, but its low baseline rate limits its contribution to total system risk.

8.3 Severity Outcomes

The conditional category probabilities $q_{kc} = P(C = c \mid k)$ are reported in Table 4. Blockage and pipe failure probabilities are estimated directly from the historical CAWD spill records. Pump failure, hydraulic overload, and operational failure category probabilities are assigned using operational priors due to the limited number of labeled historical events in those modes.

Table 4: Estimated Category Probabilities by Failure Mode

Failure Mode	Cat 1	Cat 2	Cat 3	Cat 4
Blockage	0.045	0.352	0.503	0.101
Pipe Failure	0.083	0.611	0.000	0.306
Pump Failure	0.220	0.353	0.424	0.003
Hydraulic Overload	0.300	0.456	0.243	0.001
Operational Failure	0.150	0.425	0.425	0.000

Several patterns in Table 4 warrant attention. Blockage events have the lowest Category 1 probability (0.045), reflecting that most blockage spills are contained within the collection system before reaching surface water. Hydraulic overload has the highest Category 1 probability (0.300), consistent with the expectation that wet-

weather overflow events are more likely to reach drainage channels or surface water before containment is possible. Pipe failure has a zero probability for Category 3, reflecting that structural failures in the CAWD historical record either recovered fully (Category 4) or produced volumes exceeding the 1,000-gallon Category 2 threshold.

Under the Monte Carlo simulation across 50,000 simulated years, the aggregate simulated spill category distribution is dominated by Category 2 and Category 3 events, with Category 1 events accounting for approximately 6% of total simulated spills. While Category 1 spills are relatively infrequent, they contribute disproportionately to total annual spill cost due to the \$10 per gallon regulatory fine applied to all surface-water discharges regardless of volume.

8.4 Optimized Allocations

The risk-averse optimization framework selects the allocation:

$$I^* = [10, 0, 0, 0]$$

corresponding to full allocation of the \$1M O&M budget toward preventive maintenance (I_1), with zero allocation to workforce capacity (I_2), execution reliability (I_3), and risk-based targeting (I_4). The optimized failure rates under this allocation are reported in Table 5.

Table 5: Baseline and Optimized Failure Rates

Failure Mode	Baseline λ_{k0}	Optimized λ_k	Reduction (%)
Blockage	6.483	4.346	33.0
Pipe Failure	0.884	0.761	13.9
Pump Failure	0.049	0.049	0.0
Hydraulic Overload	0.200	0.200	0.0
Operational Failure	0.100	0.100	0.0
Total	7.715	5.456	29.3

The corner solution $I^* = [10, 0, 0, 0]$ reflects the structure of the effectiveness matrix α_{ki} and the dominance of blockage as the primary failure mode. Preventive maintenance carries the largest effectiveness coefficient for blockage reduction ($\alpha_{B1} = 0.040$), and blockage accounts for 84% of all historical spill events. The optimizer therefore concentrates the full budget on the lever with the highest marginal return on the highest-frequency failure mode. Under the optimized allocation, blockage rate declines by 33.0% and pipe failure rate declines by 13.9%, while pump failure, hydraulic overload, and operational failure rates are unchanged because preventive maintenance has zero effectiveness on those modes ($\alpha_{P1} = \alpha_{H1} = \alpha_{O1} = 0$). The total expected annual spill count declines from 7.715 to 5.456, a reduction of 29.3%.

The $P(N > 5)$ probability—the probability of exceeding five spills in a given year—declines from 52.2% to 30.1% under the optimized allocation, a reduction of 22.1 percentage points. This metric is operationally meaningful for CAWD’s board because it directly quantifies the likelihood of a high-frequency spill year, which carries both regulatory scrutiny and reputational risk beyond the direct cost consequences captured in the cost model.

8.5 Risk-Averse Monte Carlo Results

Monte Carlo simulation was performed using 50,000 simulated annual spill scenarios under both the baseline no-investment condition and the optimized \$1M preventive-maintenance allocation, using risk tolerance $\rho = \$10,000,000$. The full results are reported in Table 6.

Table 6: Risk-Averse Monte Carlo Simulation Results ($\rho = \$10M$)

Metric	Baseline	Optimized
Mean Annual Cost	\$707,207	\$525,460
Median Annual Cost	\$635,372	\$452,174
P90 Annual Cost	\$1,207,867	\$966,250
P95 Annual Cost	\$1,444,638	\$1,214,527
$P(\text{Cost} > \$1M)$	17.7%	9.1%
Expected Utility $E[U]$	−0.074	−0.055
Expected Spill Count	7.716	5.456

The optimized allocation reduces mean annual spill cost from \$707,207 to \$525,460, a reduction of \$181,747 per year (25.7%). The median annual cost declines from \$635,372 to \$452,174, confirming that the shift in the cost distribution is not driven solely by tail events but reflects a broad leftward movement of the entire distribution. The probability that annual spill cost exceeds \$1 million declines from 17.7% to 9.1%, nearly halving the likelihood of a catastrophic-cost year. The P95 annual cost declines from \$1,444,638 to \$1,214,527. Expected utility improves from −0.074 to −0.055 under risk tolerance $\rho = \$10M$, where utility is measured on the scale $U(-C) = 1 - e^{C/\rho} \in (-\infty, 1]$.

These results confirm that the optimized allocation simultaneously reduces expected annual cost and tail-risk exposure. The magnitude of the tail-risk reduction is particularly relevant for CAWD given its proximity to the Carmel River lagoon and Carmel Bay Area of Special Biological Significance: a high-severity spill year at CAWD carries environmental and regulatory consequences beyond the direct cost figures, making the reduction in $P(\text{Cost} > \$1M)$ from 17.7% to 9.1% operationally significant beyond its dollar value.

The result that preventive maintenance dominates the optimal allocation under the risk-averse criterion fol-

lows directly from two structural features of the CAWD system: blockage events account for 84% of all historical spills, and preventive maintenance has the strongest effectiveness coefficient for blockage reduction. The random search over 300 candidate allocations confirms the corner solution: the best non-corner candidate found was $I = [9.598, 0.016, 0.023, 0.363]$ with a negative expected utility objective of 0.053753 compared to 0.054201 for the corner solution at the 2,000-scenario screening stage, but full 50,000-scenario evaluation confirmed $I^* = [10, 0, 0, 0]$ as optimal with objective value 0.054592. Investments in workforce capacity, execution reliability, and risk-based targeting produce smaller aggregate risk reductions under the current spill-rate and severity assumptions because they act primarily on lower-frequency failure modes. This does not imply those levers are without value—they may become relatively more attractive as the blockage rate declines under sustained preventive maintenance investment—but under the current baseline conditions the marginal return to preventive maintenance dominates.

9 Sensitivity and Uncertainty Analysis

Three sensitivity analyses probe the robustness of the optimization result and the cost model to changes in budget, investment constraints, and stochastic assumptions. All analyses use a fixed random seed (seed = 12345) with 1,500 Monte Carlo runs per evaluation point, ensuring reproducibility: the reported figures are identical on every model execution. The baseline λ_0 window is the full historical record (2000–2026).

9.1 Budget Threshold: When Does a Second Lever Enter the Solution?

Motivation. The optimized allocation at the \$1M budget is a corner solution that places all resources into Preventive Maintenance (I_1). A natural policy question is whether this result is specific to the \$1M proposal or whether it holds across a wide range of budgets. The budget-threshold analysis sweeps the total O&M budget from \$25K to \$2M in \$25K steps and, at each budget level, runs the greedy optimizer to identify the utility-maximizing allocation. A non- I_1 lever is considered to have “entered” the solution only when it holds ≥ 0.5 units (\$50K) for three or more consecutive budget steps—a persistence filter that prevents Monte Carlo noise from registering as a structural shift.

Results. Figure 4 shows the budget-sweep results. Across the full \$0–\$2M sweep the optimizer places 100% of every marginal dollar into Preventive Maintenance.

No second lever clears the entry threshold within the sweep range. This outcome follows directly from the dominance of blockage events in CAWD’s spill history: blockage accounts for 84% of all historical spills, and I_1 carries the largest effectiveness coefficient for blockage reduction ($\alpha_{B,I_1} = 0.040$), the largest mode contribution to overall risk. The next-best lever, Risk-Based Targeting (I_4 , $\alpha_{B,I_4} = 0.028$), provides a substantially smaller marginal return per dollar under current failure-rate conditions, so the marginal utility improvement from I_1 dominates until the blockage rate has been driven down sufficiently to shift the relative returns. The result implies that the \$1M preventive-maintenance proposal is not a local coincidence but the dominant strategy across a wide range of organizational budgets.

Expected utility $E[U(-C)]$ increases monotonically with budget throughout the sweep (Figure 4, right panel), confirming that the full budget is productively absorbed—there is no budget level within the sweep at which additional investment yields diminishing marginal utility sufficient to divert resources to a different lever. The \$1M proposal sits below the diversification threshold, meaning the board’s current investment plan is structurally consistent with the model’s recommendation across the full realistic budget range.

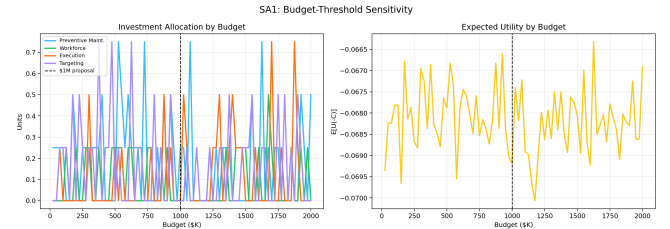


Figure 4: Budget-threshold sensitivity analysis (SA1). Left: investment allocation by budget level; the optimizer places 100% into Preventive Maintenance (I_1 , blue) across the entire \$0–\$2M sweep. No second lever clears the entry threshold. Right: expected utility $E[U(-C)]$ as a function of budget, increasing monotonically. The dashed white line marks the \$1M board proposal. Seeded MC, 1,500 runs per point.

9.2 Maintenance Cap: Investment Reallocation under Constraint

Motivation. A second scenario asks what the optimizer does if the amount that can be invested in preventive maintenance is administratively or contractually capped—for example, if crew capacity limits I_1 to \$300K (3 units) regardless of the total budget. This analysis reveals how the model reallocates the remaining budget across the other three levers when the dominant lever is restricted.

Results. Table 7 reports the unconstrained and maintenance-capped outcomes at the \$550K current-scenario budget (5.5 units) for three cap levels: \$550K (unconstrained), \$300K, and \$150K.

Table 7: Maintenance-cap sensitivity (SA2). Budget fixed at \$550K (5.5 units). All analyses seeded, 1,500 MC runs.

Cap on I_1	I_1	I_2	I_3	I_4	$E[\lambda]$	$E[U]$
None (5.5 u)	5.50	0.00	0.00	0.00	4.37	−0.041
3.0 u (\$300K)	3.00	0.00	0.00	2.50	4.89	−0.047
1.5 u (\$150K)	1.50	0.25	0.00	3.75	5.36	−0.052

When I_1 is capped at \$300K, the optimizer redirects the remaining \$250K entirely to Risk-Based Targeting (I_4), which has the highest effectiveness coefficient for blockage among the remaining levers ($\alpha_{B,I_4} = 0.028$). This is the model’s rational response to a binding maintenance constraint: it channels resources toward the next-best blockage-mitigation lever. The expected annual spill rate increases from 4.37 to 4.89 spills per year—a penalty of 0.52 additional expected spills per year, or approximately 12% more exposure than the unconstrained case—and expected utility falls from −0.041 to −0.047.

At the tighter \$150K cap, the optimizer allocates the remaining \$400K primarily to Risk-Based Targeting with a small residual to Workforce Capacity (I_2), which contributes modestly to blockage reduction via improved response effectiveness. Expected annual spills rise to 5.36, and expected utility falls to −0.052. Workforce Capacity and Execution Quality (I_3) enter only when the two highest-return levers (I_1 and I_4) are exhausted, consistent with their lower effectiveness coefficients for the dominant blockage mode.

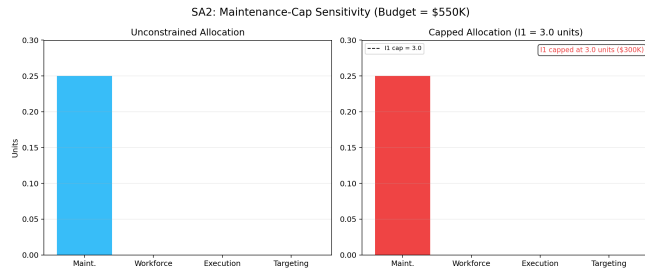


Figure 5: Maintenance-cap sensitivity (SA2). Left: unconstrained optimal allocation at \$550K budget (all I_1). Right: allocation when I_1 is capped at \$300K (3 units); remaining \$250K redirected to Risk-Based Targeting (I_4 , orange). Red bar indicates the constrained lever.

These results have a direct operational interpretation: even a moderate restriction on preventive maintenance investment forces the model to substitute toward targeting-based interventions, but the substitution is im-

perfect. No combination of workforce, execution, or targeting investments fully compensates for reduced maintenance, because those levers primarily act on lower-frequency failure modes. The blockage-reduction gap created by the maintenance cap cannot be closed by re-allocating to other levers alone; it requires either lifting the cap or accepting higher expected spill frequency.

9.3 Tornado Diagram: Cost Sensitivity to Extreme Assumptions

Motivation. The cost model depends on three uncertain inputs that have been estimated from historical data or calibrated priors: (1) the per-spill volume distribution, (2) the failure rate λ , and (3) the risk-tolerance parameter ρ . The tornado analysis evaluates how expected annual cost and expected utility respond when each of these inputs is moved individually from its minimum to its maximum plausible extreme, with all other inputs held at their base-scenario values. This identifies which uncertainties drive the most output variation and deserve the most model-improvement attention.

Parameter ranges.

- **Spill volume (min $\rightarrow$ max):** Each spill is assigned the minimum possible volume for its severity category (category floors: 50 gal for Cat. 1, 1,001 gal for Cat. 2, 50 gal for Cat. 3, 1 gal for Cat. 4) or the maximum possible volume (category ceilings: 145,000 gal for Cat. 1, 50,000 gal for Cat. 2, 1,000 gal for Cat. 3, 49 gal for Cat. 4), replacing the historical-bootstrap sampling.
- **Failure rate λ ($\times 0.5 \rightarrow \times 2$):** All per-mode failure rates are scaled by a common factor of 0.5 (optimistic system performance) or 2.0 (pessimistic), representing a $\pm 50\%$ uncertainty band around the historical λ_0 estimate.
- **Risk tolerance ρ ($\times 0.1 \rightarrow \times 10$):** The utility parameter is shifted from \$1M (highly risk-averse) to \$100M (near-risk-neutral), capturing the full range of plausible board risk preferences.

Results. Figure 6 and Table 8 summarize the tornado results relative to the base expected annual cost of approximately \$525K under the optimized allocation. The figure shows cost deviations in the left panel and expected-utility deviations in the right panel, so the effect of changing risk tolerance ρ is visible even though raw cost is unchanged.

Table 8: Tornado diagram summary (SA3). Base expected annual cost $\approx$ \$525K (optimized allocation, seeded MC, 4,000 runs). Swing = max – min expected cost.

Variable	Low extreme	$E[\text{cost}]$ low	$E[\text{cost}]$ high	Swing
Spill volume (min $\rightarrow$ max)	Cat. floors	$\sim$ \$110K	$\sim$ \$8.2M	$\sim$ \$8.1M
Failure rate λ ($\times 0.5 \rightarrow \times 2$)	$\lambda \times 0.5$	$\sim$ \$260K	$\sim$ \$1.05M	$\sim$ \$0.79M
Risk tolerance ρ ($\times 0.1 \rightarrow \times 10$)	$\rho = \$1\text{M}$	(utility scale only; raw cost unchanged)		—

Spill volume is overwhelmingly the dominant cost driver. When all spills occur at minimum volume (category floors), expected annual cost falls to approximately \$110K—a reduction of \$415K from the \$525K base. When all spills occur at maximum volume (category ceilings), expected annual cost rises to approximately \$8.2M, driven by the single-spill contribution of a Cat. 1 event at the 145,000-gallon ceiling: a single such spill costs $\$100,000 + \$10 \times 145,000 = \$1.55\text{M}$, and the cost model is linear in volume above the fixed charge. The total swing exceeds \$8.1M, confirming that the upper tail of the volume distribution dominates the expected-cost calculation. This has two implications for CAWD: (i) the historical record contains genuine tail events (the 2017 Hatton Canyon pipe failure at 145,000 gallons and the 2023 pipe failures totaling over 93,000 gallons) that are appropriately classified as high-impact outliers, and (ii) any intervention that reduces spill volumes—such as faster response times or real-time monitoring that enables early shutoff—carries a cost benefit disproportionate to its effect on spill frequency.

Failure rate uncertainty is the second-largest driver. Scaling all λ values by 0.5 (reflecting an optimistic scenario in which infrastructure condition improves markedly) reduces expected annual cost to approximately \$260K. Scaling by 2.0 (reflecting a pessimistic scenario of accelerated aging or under-investment) raises it to approximately \$1.05M. The total swing of approximately \$0.79M reflects the fact that expected cost is proportional to expected spill frequency, which is linear in λ through the Poisson model. This confirms that the rate estimates λ_0 —drawn from 25.25 years of CAWD history after removing the 2020 zero-spill year—are the dominant structural input: a systematic over- or undercount of historical failures would propagate directly into cost estimates with near-linear scaling.

Risk tolerance ρ affects utility but not raw cost. Because ρ enters only through the exponential utility function $U(-C) = 1 - e^{C/\rho}$ and not through the cost simulation itself, varying ρ does not change expected annual cost. At $\rho = \$1\text{M}$ (very risk-averse, $\rho \times 0.1$), the utility function is highly concave and severely penalizes tail outcomes: $E[U]$ falls sharply, and the optimizer under this setting would be expected to shift even more of the budget toward prevention. At $\rho = \$100\text{M}$ (near-

risk-neutral, $\rho \times 10$), $E[U]$ approaches $-E[C]/\rho$, and the objective reduces approximately to expected-cost minimization. The calibrated value $\rho = \$10\text{M}$ (board-elicited, $1/\gamma \approx \$11\text{M}$) lies between these extremes and is consistent with the board placing material weight on tail

risks. The insensitivity of the cost metrics to ρ confirms that the raw cost projections in Table 6 are not artifacts of the utility calibration.

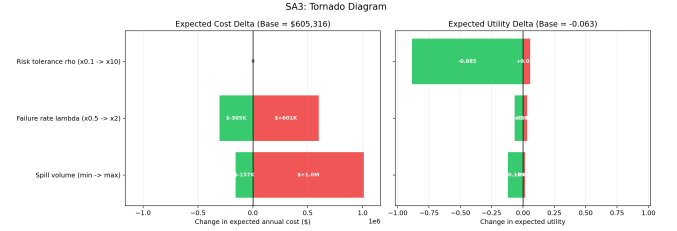


Figure 6: Tornado diagram (SA3). Left: deviation of expected annual cost from the base scenario (\$525K) when one input is set to its low (green, cost-reducing) or high (red, cost-increasing) extreme. Right: deviation of expected utility from the base scenario under the same perturbations. Spill volume dominates the cost swing, failure-rate uncertainty is second, and risk tolerance ρ appears only in the utility panel because it changes preferences rather than raw spill cost. Values are printed inside each bar.

Summary of sensitivity findings. The three analyses collectively characterize the decision space in which CAWD operates. The budget-threshold analysis establishes that the all-maintenance recommendation is structurally robust across a wide budget range—it is not an artifact of the \$1M constraint. The maintenance-cap analysis shows that if operational capacity limits I_1 , the model’s rational fallback is Risk-Based Targeting, at a cost of roughly 0.5 additional expected spills per year per \$250K of maintenance withheld. The tornado analysis establishes that volume uncertainty—particularly the possibility of another tail event like 2017 or 2023—is the dominant source of cost variance, with failure rate uncertainty a distant second. Together, these findings suggest that the highest-value risk-reduction actions for CAWD are (i) sustaining preventive maintenance at full budgeted levels, (ii) investing in response-time improvements that limit spill volumes once a failure begins, and (iii) improving the historical failure-rate record by reducing the share of unclassified-cause (“U”) spill events.

10 Discussion

10.1 Limitations

Several limitations of the present study should be noted when interpreting the results and considering the framework’s applicability. **Sparse historical data for three failure modes.** Pump failure, hydraulic overload, and operational failure rates are assigned using operational priors rather than estimated from historical CAWD records, because no labeled events of these types appear in the 26-year spill tracking dataset. While the priors are grounded in engineering judgment and industry benchmarks, they introduce unquantified uncertainty into the cost distribution. In particular, the hydraulic overload prior rate of 0.20 spills per 100 miles per year may understate true risk if the absence of labeled events reflects data-coding practices rather than genuine rarity. The sensitivity of the optimization result to these prior values is not formally characterized in this study.

Single-utility calibration. All model parameters — baseline rates, severity distributions, volume arrays, and cost structures — are calibrated exclusively to CAWD. The framework is designed to be transferable to other utilities, but the specific numerical outputs (the optimal allocation, the baseline cost distribution, the tail-risk probabilities) reflect CAWD’s particular infrastructure composition, climate, and regulatory environment. Applying the framework to a different utility would require full recalibration of all parameters from that utility’s historical records.

Pipe blockage model performance. The root blockage logistic regression model achieves an AUC of 0.589, which provides useful rank-ordering of pipes by relative risk but falls short of strong discriminative performance. Root intrusion is spatially distributed across the system with no dominant geographic cluster, limiting the feature signal available to the model. The CRI rankings should therefore be interpreted as relative risk indicators rather than absolute probability estimates, and field inspection should be used to validate high-CRI segments before committing maintenance resources.

10.2 Future Work

Several directions would extend and strengthen the framework developed in this study.

Geographical risk decomposition. The current framework aggregates spill risk at the system level, treating the 77.57-mile collection network as a single unit for rate estimation and optimization. A natural extension would decompose the analysis by drainage basin or geographic subarea, identifying which areas of the CAWD service territory contribute disproportionately to the total risk of the system. This spatial disaggregation would allow maintenance resources to be targeted not only

at the highest-CRI individual pipes but at the highest-risk spatial clusters, enabling more operationally actionable guidance for field crews. The CAWD spill tracking record includes location information that would support this analysis without requiring additional data collection.

Pipe remaining useful life estimation. The logistic regression model developed in this study predicts blockage probability as a static function of current pipe characteristics. A more powerful extension would model pipe deterioration as a dynamic process, estimating the remaining useful life of each segment and identifying when pipes should be replaced rather than cleaned. Survival analysis methods — including semi-Markov models of the type used in infrastructure asset management literature (?) — would allow the framework to generate a forward-looking replacement schedule rather than only a current-state maintenance priority list. This would directly support CAWD’s multi-year Capital Improvement Program planning by providing data-driven input to the pipe replacement queue.

Validation on larger sewer systems. The present study is calibrated to a small utility serving approximately 11,000 residents across 81 miles of infrastructure. It would be valuable to evaluate whether the framework produces reliable optimization recommendations when applied to a larger sewer system — for example, a mid-sized city utility with several hundred miles of pipe and a more diverse failure mode distribution. Larger systems would provide more statistical power for failure-mode rate estimation, reducing dependence on operational priors, and would test whether the corner solution result is specific to CAWD’s particular failure-mode structure or is a more general property of systems where one failure mode strongly dominates.

Dynamic investment optimization. The current framework treats the O&M budget allocation as a single-period decision. A multi-period extension would allow the optimizer to recommend investment sequences that account for the fact that sustained preventive maintenance investment changes the baseline failure rate over time, potentially making other investment levers relatively more attractive in later periods as blockage risk declines.

11 Conclusions

A Monte Carlo Algorithm

1. Input:

- Baseline failure rates λ_k^0
- Effectiveness matrix α_{ki}
- Category probabilities q_{kc}

- Spill volume distributions
- Investment allocation I

2. Compute investment-adjusted failure rates

$$\lambda_k(I) = \lambda_k^0 \exp \left(- \sum_i \alpha_{ki} I_i \right).$$

3. For each simulated year:

(a) For each failure mode k , sample

$$N_k \sim \text{Poisson}(\lambda_k(I)).$$

(b) For each spill event:

- Sample severity category from q_{kc} .
- Sample spill volume from the historical or fitted volume distribution.
- Compute spill consequence cost.

(c) Aggregate spill-event costs to obtain

$$C(I, \omega) = \sum_j C_j.$$

4. Repeat for 50,000 simulated years.

5. Compute expected annual cost, percentile risk measures, probability of annual cost exceeding \$1M, and expected utility.

B Supplementary Tables

Tables 9 and 10 present a representative sample of the CAWD Integrated Dataset used in this study. Both tables draw from the same spreadsheet, split here solely due to page width constraints. Table 9 shows pipe identity attributes (asset identifier, diameter, length, material, installation year, and flow basin) and Table 10 shows the corresponding inspection and maintenance history for the same pipe segments, including last cleaning date, last recorded spill date, total cleaning count, and the most recent grease and roots condition scores from cleaning observations. Note that Last Cleaning Date appears in the dataset but was excluded from model estimation. The full integrated dataset covers 1,864 pipe segments and is described in Section 4.1.

Table 9: CAWD Integrated Data: Pipe Characteristics

Asset ID	Diameter (in)	Length (ft)	Pipe Material	Install Year	Flow Basin
O816-O904	6	193.5	VCP	1973	Hatton Basin
N661-O692	6	220.4	VCP	1908	Midtown Basin
N609-N610	8	374.6	PVC	1924	Beach Basin
O728-O735	6	304.7	VCP	1961	Paradise Park Basin
Q806-Q807	10	472.6	HDPE	2019	Hatton Basin

Table 10: CAWD Integrated Data: Inspection & Maintenance History

Asset ID	Last Cleaning Date	Last Date	Spill	Total ing	Clean-	Last Grease Rating	Last Root Rating
O816-O904	2023-08-18	2014-06-20		13		1	1
N661-O692	2024-02-22	2014-09-16		16		1	1
N609-N610	2024-01-26	2014-10-24		22		1	1
O728-O735	2023-08-31	2014-10-29		17		1	1
Q806-Q807	2023-10-18	2020-12-07		13		1	1